

Modular Symmetry Fields in Finite Quantum Theory: A Finite-Field Formulation with Conjectural Physical Consequences

Brahimi, Mahdi Tahar

Department of Mathematics

University of Mohamed Boudiaf, Msila, Algeria

mahditahar.brahimi@univ-msila.dz

ABSTRACT.

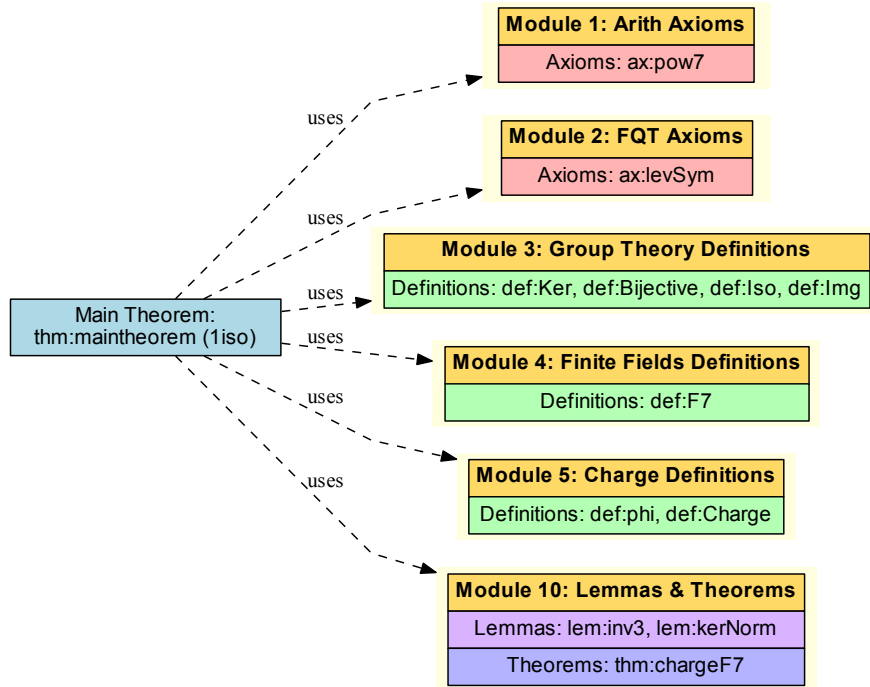
Framework := $\text{FQT} \wedge F_p \wedge F_{p^2} \wedge \text{ModIR} \wedge \text{dS} \wedge \text{Contraction}$

Structure₁ := $R_p = \mathbb{Z}/p\mathbb{Z} \wedge \text{GF}_7 \subseteq R_p \wedge (\forall (n : \mathbb{Z}), \varphi(n/3) = 5n \bmod 7)$

Structure₂ := $\text{ModIR}(\mathfrak{sl}(2), F_p) \wedge (\forall (\lambda : \text{Fin } p), \dim L(\lambda) = \lambda + 1) \wedge \text{Steinberg}(L(p - 1))$

Structure₃ := $\text{Gal}(\text{GF}_{2^{615}}/\text{GF}_2) \cong \mathbb{Z}/615\mathbb{Z} \wedge \tau(615) = 8$

Contraction := $(R_p \xrightarrow{p \rightarrow \infty} \mathbb{Z}) \wedge (\text{dS} \xrightarrow{R \rightarrow \infty} \text{Poincaré})$



1 Axioms

1.1 Arith

id	type
1. $\mathbf{Ax}_{\text{addMon}}$	$\forall (a \ b \ c : \mathbb{R}), a = b \rightarrow a + c = b + c$
2. $\mathbf{Ax}_{\text{mulMon}}$	$\forall (a \ b \ c : \mathbb{R}), a = b \rightarrow a \cdot c = b \cdot c$
3. $\mathbf{Ax}_{\text{arith}}$	$\forall (a \ b \ c \ d : \mathbb{Q}), (a = c \wedge b = d) \rightarrow (a + b = c + d \wedge a \cdot b = c \cdot d)$
4. $\mathbf{Ax}_{\text{pow}}$	$\forall (a : \mathbb{N})(m \ n : \mathbb{N}), a^{m+n} = a^m \cdot a^n \wedge a^{m \cdot n} = (a^m)^n$
5. $\mathbf{Ax}_{\text{pow7}}$	$2^3 = 8 \wedge 2^3 - 1 = 7 \wedge 3 \cdot 5 \equiv 1 \pmod{7}$

1.2 FQT

id	type
6. $\mathbf{Ax}_{\text{sym}}$	$\forall (\mathcal{A} : \text{Type})(n : \mathbb{N})(A : \text{Fin}n \rightarrow \mathcal{A})(c : \text{Fin}n \times \text{Fin}n \times \text{Fin}n \rightarrow \mathbb{Z}),$ $(\forall (j \ k : \text{Fin}n), [A_j, A_k] = \sum_{l : \text{Fin}n} c(j, k, l) \cdot A_l) \rightarrow \text{SymDef}(\mathcal{A}, A, c)$
7. $\mathbf{Ax}_{\text{dS}}$	$\forall (M : \text{Fin}5 \times \text{Fin}5 \rightarrow \mathcal{A})(\eta : \text{Fin}5 \rightarrow \mathbb{Z}),$ $(\forall (a \ b \ c \ d : \text{Fin}5), [M_{ab}, M_{cd}] = -i(\eta_{ac}M_{bd} + \eta_{bd}M_{ac} - \eta_{ad}M_{bc} - \eta_{bc}M_{ad}))$ $\rightarrow \text{dSAlg}(M, \eta)$
8. $\mathbf{Ax}_{\text{cont}}$	$\forall (R : \mathbb{R}_{>0})(M : \text{Fin}5 \times \text{Fin}5 \rightarrow \mathcal{A}), P_\mu = M_{4\mu}/R \rightarrow_{\text{formal}} \text{Poincaré}(P, M)$
9. $\mathbf{Ax}_{\text{finite}}$	$\forall (p : \mathbb{N}), \text{Prime}(p) \rightarrow (\text{FQT} = \text{QM}(F_p) \wedge \lim_{p \rightarrow \infty} (\text{FQT}) = \text{QM}(\mathbb{C}))$
10. $\mathbf{Ax}_{\text{modular}}$	$\forall (p : \mathbb{N}), \text{Prime}(p) \rightarrow (\text{FinDim}(\text{ModIR}(\mathfrak{sl}(2), F_p)) \wedge \text{FinDim}(\text{ModIR}(\mathfrak{so}(1, 4), F_p)))$
11. $\mathbf{Ax}_{\text{singleton}}$	$\text{Conj}(\forall (P : \text{Particle}), \text{Elementary}(P) \leftrightarrow \text{DiracSingleton}(P))$
12. $\mathbf{Ax}_{\text{dyn}}$	$\forall (H : \mathcal{A}), \text{Hamiltonian}(H) \rightarrow H \in U(\text{SymAlg})$
13. $\mathbf{Ax}_{\text{arrow}}$	$\text{FQT} \rightarrow \text{QM}(\mathbb{C}) \rightarrow \text{Classical}$
14. $\mathbf{Ax}_{p \text{ large}}$	$p \gg 1 \wedge \ln p \gg 1$
15. $\mathbf{Ax}_{\text{Ham}}$	$\forall (H : M_N(F_p)), \text{Hamiltonian}(H)$
16. $\mathbf{Ax}_{\text{Time}}$	$\forall (U : \mathbb{Z} \rightarrow \text{GL}_N(F_p))(g : \text{GL}_N(F_p)), U(1) = g \wedge \text{Finite}(\text{order}(g)) \rightarrow \text{TimeEvo}(U)$
17. $\mathbf{Ax}_{\text{Obs}}$	$\forall (A : M_N(F_p)), \text{Eigen}(A) \subseteq \text{SplittingField}(A)$

1.3 LevAssumptions

id	type
18. $\mathbf{Ax}_{\text{PhysNonPhys}}$	$\forall (n : \mathbb{N}^4), \text{Physical}(n) \leftrightarrow \neg \text{Physical}(\bar{n})$ (Physical-nonphysical states assumption of Lev, §8.5)
19. $\mathbf{Ax}_{\text{StdInterp}}$	$\forall (\mathcal{A} : \text{Type})(N : \mathbb{N}),$ $\text{Rep}(\mathcal{A}, N) \rightarrow (\text{FullSpace}(\mathcal{A}) \wedge \text{TensorProduct}(\mathcal{A}, N))$ (Standard-Interpretation Requirements of Lev, §8.5)
20. $\mathbf{Ax}_{\text{Gödel}}$	$\neg \exists (\mathbf{Ax} : \text{AxiomSet}), \text{Complete}(\mathbf{Ax}) \wedge \text{Consistent}(\mathbf{Ax})$

2 Definitions

2.1 GroupTheory

id	type
1. Hom	$\text{Hom} : \forall (G : \text{Type})[\text{Group}(G)](H : \text{Type})[\text{Group}(H)], (G \rightarrow H) \rightarrow \text{Prop}$
2. ker	$\text{ker} : \forall (G \ H : \text{Type})[\text{Group}(G)][\text{Group}(H)], (G \rightarrow H) \rightarrow \text{Subgroup}(G)$
3. Img	$\text{Img} : \forall (G \ H : \text{Type}), (G \rightarrow H) \rightarrow \text{Set}(H)$
4. Bijective	$\text{Bijective} : \forall (\alpha \ \beta : \text{Type}), (\alpha \rightarrow \beta) \rightarrow \text{Prop}$

id	type
5. Injective	$\text{Injective} : \forall (\alpha \beta : \text{Type}), (\alpha \rightarrow \beta) \rightarrow \text{Prop}$
6. Surjective	$\text{Surjective} : \forall (\alpha \beta : \text{Type}), (\alpha \rightarrow \beta) \rightarrow \text{Prop}$
7. Iso	$\text{Iso} : \forall (G H : \text{Type}), \text{Bijective}(G, H) \rightarrow \text{Prop}$
8. Cyc	$\text{Cyc} : \forall (G : \text{Type})[\text{Group}(G)], \text{Prop}$
9. Sub	$\text{Sub} : \forall (G : \text{Type}), \text{Set}(G) \rightarrow \text{Prop}$
10. order	$\text{order} : \forall (G : \text{Type})[\text{Group}(G)], G \rightarrow \mathbb{N}$

2.2 FiniteFields

id	type
11. F_p	$F_p : \mathbb{N} \rightarrow \text{Type} \quad (p \mapsto \mathbb{Z}/p\mathbb{Z})$
12. GF_7	$\text{GF}_7 : \text{Type} := \mathbb{Z}/7\mathbb{Z}$
13. GF_8	$\text{GF}_8 : \text{Type} := \text{GF}_{2^3}$
14. $\text{GF}_8^\times$	$\text{GF}_8^\times : \text{Subgroup}(\text{GF}_8) := \lambda(x : \text{GF}_8), x \neq 0$
15. Subfield	$\text{Subfield} : \mathbb{N} \rightarrow \mathbb{N} \rightarrow \text{Prop} \quad ((m, k) \mapsto \text{GF}_{2^m} \hookrightarrow \text{GF}_{2^k})$
16. FF	$\text{FF} : \mathbb{N} \rightarrow \text{Type} \quad (k \mapsto \text{GF}_{2^k})$
17. Gal	$\text{Gal} : \forall (k : \mathbb{N}), \text{Aut}(\text{FF}(k)/\text{GF}_2)$
18. τ	$\tau : \mathbb{N} \rightarrow \mathbb{N} \quad (n \mapsto \#\{d : \mathbb{N} \mid \text{Div}(d, n)\})$
19. CharPoly	$\text{CharPoly} : \forall (N : \mathbb{N}) (A : M_N(F_p)), F_p[x]$
20. SplittingField	$\text{SplittingField} : \forall (p N : \mathbb{N}) (A : M_N(F_p)), \text{Field}(\text{GF}_{p^m}) \wedge m = \text{lcm}(d_1, \dots, d_k)$
21. Eigen	$\text{Eigen} : \forall (p N : \mathbb{N}) (A : M_N(F_p)), \text{Roots}(\text{CharPoly}(A)) \subseteq \text{GF}_{p^m}$

2.3 Charge

id	type
22. Charge	$\text{Charge} : \text{Set}(\mathbb{Q}) := \lambda(q : \mathbb{Q}), \exists (n : \mathbb{Z}), q = n/3 \wedge q \leq 1$
23. φ	$\varphi : \mathbb{Z} \rightarrow \text{GF}_7 := \lambda(n : \mathbb{Z}), (5 \cdot n) \bmod 7$
24. φ_H	$\varphi_H : \text{Hom}((1/3)\mathbb{Z}, \text{GF}_7, \varphi)$
25. ChargeIsoF7	$\text{ChargeIsoF7} := \varphi_H \wedge \text{Bijective}(\text{Charge}, \text{GF}_7, \varphi)$

2.4 ModularRep

id	type
26. $\mathfrak{sl}(2)(F_p)$	$\mathfrak{sl}(2)(F_p) : \text{Type} := \lambda(X : M_2(F_p)), \text{tr}(X) = 0$
27. $u(\mathfrak{sl}(2))$	$u(\mathfrak{sl}(2)) := U(\mathfrak{sl}(2))/\langle x^p - x^{[p]} : x : \mathfrak{sl}(2) \rangle$
28. ModIR	$\text{ModIR} : \forall (\mathfrak{g} : \text{LieAlg})(k : \text{Type}), \text{Type}$
29. L	$L : \text{Finp} \rightarrow \text{ModIR}(u(\mathfrak{sl}(2)))$
30. St	$\text{St} : \text{ModIR}(u(\mathfrak{sl}(2))) := L(p - 1)$
31. L_{gen}	$L_{\text{gen}} : \mathbb{N} \rightarrow \text{ModIR}(u(\mathfrak{sl}(2))), \lambda = \lambda_0 + p\lambda_1 \rightarrow L_{\text{gen}}(\lambda) = L(\lambda_0) \otimes L(\lambda_1)^{(1)}$
32. WeightSet	$\text{WeightSet} : \forall (\lambda : \text{Finp}), \text{Set}(\mathbb{Z}) := \{\lambda, \lambda - 2, \dots, -\lambda\}$
33. TensProd	$\text{TensProd} : \forall (\lambda \mu : \text{Finp}), L(\lambda) \otimes L(\mu) \rightarrow \text{Prop}$
34. BrauerChar	$\text{BrauerChar} : \forall (M : \text{ModIR}(u(\mathfrak{sl}(2)))) , \text{ClassFn}(G_p)$
35. SimpleMod	$\text{SimpleMod} : \text{ModIR}(u(\mathfrak{sl}(2))) \rightarrow \text{Prop}$

2.5 Contraction

id	type
36. h	$h : \mathbb{N} \rightarrow \mathbb{N} := \lambda(p : \mathbb{N}), \lfloor p/2 \rfloor$
37. PolyData	$\text{PolyData} : \forall(f : \mathbb{Z}[x_1, \dots, x_n]), (M : \mathbb{N}) \times (C : \mathbb{N}) \times (d : \mathbb{N}) \times (B : \mathbb{N})$
38. Intermediate	$\text{Intermediate} : \forall(f : \mathbb{Z}[x_1, \dots, x_n])(\text{eval})(v : \mathbb{Z}), \text{Prop}$
39. IntermediateBound	$\text{IntermediateBound} : \forall(f M C B d), \max_{\text{eval}}(v) \leq M \cdot C \cdot B^d$
40. Faithful	$\text{Faithful} : \forall(f M C B d p), (M \cdot C \cdot B^d < h(p)) \rightarrow (\text{eval}_{F_p}(f) = \text{red}_p(\text{eval}_{\mathbb{Z}}(f)))$

2.6 Dynamics

id	type
41. TimeEvo	$\text{TimeEvo} : \mathbb{Z} \rightarrow \text{GL}_N(F_p)$
42. order_G	$\text{order}_G : \text{GL}_N(F_p) \rightarrow \mathbb{N}$
43. Periodic	$\text{Periodic} : \forall(U : \mathbb{Z} \rightarrow \text{GL}_N(F_p))(r : \mathbb{N}), (\forall(t : \mathbb{Z}), U(t + r) = U(t))$
44. $\text{GL}_N(F_p)$	$\text{GL}_N(F_p) : \text{Type}, \# \text{GL}_N(F_p) = p^{N(N-1)/2} \prod_{i=1}^N (p^i - 1)$
45. PowerTwoScale	$\text{PowerTwoScale} : \text{Creation} \rightarrow \mathbb{N} \rightarrow \mathbb{N} \rightarrow \text{Prop}$
46. LinOrd	$\text{LinOrd} : \forall(\alpha : \text{Type}), (\alpha \rightarrow \alpha \rightarrow \text{Prop}) \rightarrow \text{Prop}$

2.7 LevDefinitions

id	type
47. GenTheory	$\forall(A B : \text{Theory})(c_0 : \mathbb{R} \cup \{\pm\infty\}),$ $(\text{FiniteParam}(A, c_0) \wedge B = \lim_{c \rightarrow c_0} A \wedge \forall(\varepsilon > 0) \exists(c) \text{Reproduce}(A, B, c, \varepsilon))$ $\rightarrow \text{MoreGeneral}(A, B)$ (Definition 1.3 of Lev)
48. ABSym	$\forall(a b : \text{Op}),$ $(\text{ACR}(b, b^*) \wedge \forall(A : \text{RepOp}), \exists(A_{ij}) (A = \sum_{i,j} A_{ij} b_i^* b_j))$ $\rightarrow \text{ABSym}(a, b)$ (Definition of the AB symmetry, Lev §8.5)
49. $\text{osp}(1, 4)$	$\forall(A B C : \text{FermOp}),$ $[A, \{B, C\}] = F(A, B) C + F(A, C) B$ (where F skew-symmetric and $F(d'_j, d''_j) = 1$) (Lev §8.10)
50. Sing	$\forall(e_0 : \text{RepSpace})(d'_j, d''_j : \text{FermOp}),$ $(d'_j e_0 = 0 \wedge d'_j d''_j e_0 = \frac{p+1}{2} e_0)$ $\rightarrow \text{Supersingleton}(e_0)$ (Lev §9.5)

3 Statements

3.1 Lemmas

id	type
1. L_{inv3}	$3 \cdot 5 \equiv 1 \pmod{7} \rightarrow 3^{-1} \equiv 5 \pmod{7}$
2. L_{Sub}	$\text{Sub}((\mathbb{Q}, +), (1/3)\mathbb{Z})$
3. L_{Norm}	$\text{Normal}(\ker(\varphi), (1/3)\mathbb{Z})$
4. L_{Uniq}	$\forall(\psi : (1/3)\mathbb{Z} \rightarrow \text{GF}_7), (\text{Hom}((1/3)\mathbb{Z}, \text{GF}_7, \psi) \wedge \psi(1/3) = 5) \rightarrow \psi = \varphi$
5. L_{Cont}	$\forall(f : \mathbb{Z}[x_1, \dots, x_n]) (\text{eval}), (\forall(v : \mathbb{Z}), \text{Intermediate}(f, \text{eval}, v) \rightarrow v < h(p)) \rightarrow$ $\text{Faithful}(f, p)$

id	type
6. $\mathbf{L}_{\text{Ferm}}$	$\forall(D : \text{FermOp})(E : \text{RepSpace}),$ $\exists(O : \text{BosOp})(c_1 \ c_2 : \mathbb{R}), \ DE = c_1 E + c_2 O E$ (Lev §8.10)

3.2 Theorems

id	type
1. $\mathbf{T}_{\text{ChargeGF7}}$	$\varphi_H \wedge \text{Bijective}(\text{Charge}, \text{GF7}, \varphi)$
2. $\mathbf{T}_{\text{Iso}}$	$\exists! (\bar{\varphi} : (1/3)\mathbb{Z}/\ker(\varphi) \rightarrow \text{GF7}), \ \text{Iso}((1/3)\mathbb{Z}/\ker(\varphi), \text{GF7}, \bar{\varphi}) \wedge \bar{\varphi} \circ \pi = \varphi$
3. $\mathbf{T}_{\text{GF8}^\times}$	$\text{Cyc}(\text{GF8}^\times) \wedge (\text{GF8}^\times, \cdot) \cong (\mathbb{Z}/7\mathbb{Z}, +) \cong (\text{GF7}, +)$
4. $\mathbf{T}_{\text{Subfield}}$	$\forall(k : \mathbb{N}), \ \#\text{Subfield}(\text{FF}(k)) = \tau(k)$
5. $\mathbf{T}_{\text{Lat}}$	$\forall(n : \mathbb{N}), \ \text{Sub}(\mathbb{Z}/n\mathbb{Z}) \cong \{d : \mathbb{N} \mid \text{Div}(d, n)\} \wedge \#\text{Sub}(\mathbb{Z}/615\mathbb{Z}) = \tau(615) = 8$
6. $\mathbf{T}_{\text{Simp}}$	$\text{SimpleMod}(u(\text{sl}(2))) = \{L(0), \dots, L(p-1)\} \wedge \forall(\lambda : \text{Finp}), \ \dim L(\lambda) = \lambda + 1$
7. $\mathbf{T}_{\text{St}}$	$\text{Proj}(L(p-1)) \wedge \text{Inj}(L(p-1))$
8. $\mathbf{T}_{\text{Wt}}$	$\forall(\lambda : \text{Finp}), \ \text{WeightSet}(L(\lambda)) = \{\lambda, \lambda-2, \dots, -\lambda\} \wedge \dim L(\lambda) = \lambda + 1$
9. $\mathbf{T}_{\text{Tens}}$	$\forall(\lambda \ \mu : \text{Finp}),$ $(\lambda + \mu \leq p-1 \rightarrow \text{TensProd}(L(\lambda), L(\mu)) \cong \bigoplus_{k=0}^{\min(\lambda, \mu)} L(\lambda + \mu - 2k)) \wedge$ $(\lambda + \mu \geq p \rightarrow \neg \text{SemiSimple}(\text{TensProd}(L(\lambda), L(\mu))))$
10. $\mathbf{T}_{\text{Char}}$	$\forall(\lambda : \text{Fin7}), \ \text{ch}(L(\lambda)) = \sum_{k=0}^{\lambda} e_{\lambda-2k}$
11. $\mathbf{T}_{\text{StDim}}$	$\dim L(p-1) = p \wedge \dim L(6) = 7$
12. $\mathbf{T}_{\text{ModCharge}}$	$\text{Bijective}(\text{WeightSet}(L(6)), \text{GF7}, \text{red7}) \wedge \text{Bijective}(\text{Charge}, \text{WeightSet}(L(6)), \lambda(n : \mathbb{Z}), 2n \bmod 7)$
13. $\mathbf{T}_{\text{Quant}}$	$\forall(f \ M \ C \ B \ d), \ (M \cdot C \cdot B^d < h(p)) \rightarrow \text{Faithful}(f, p)$
14. $\mathbf{T}_{\text{Error}}$	$\forall(M \ C \ B \ d), \ p > 2 \cdot M \cdot C \cdot B^d \rightarrow \text{Faithful}(f, p) \wedge (M \cdot C = 1 \wedge B \leq 2 \rightarrow p > 2^{d+1})$
15. $\mathbf{T}_{\text{PBound}}$	$\forall(N \ d \ B \ M \ C), \ (M \cdot C \cdot B^d < h(p)) \rightarrow \text{Faithful}(f, p)$
16. $\mathbf{T}_{\text{FinSpec}}$	$\forall(N : \mathbb{N}) \ (A : M_N(F_p)), \ \text{CharPoly}(A) = \prod_{i=1}^k \chi_i^{e_i} \rightarrow$ $(\sum_i \deg(\chi_i) \leq N) \wedge \text{Eigen}(A) \subseteq \text{GF}_p^m \wedge m = \text{lcm}(d_1, \dots, d_k)$ $\wedge m \mid N!$
17. $\mathbf{T}_{\text{Period}}$	$\forall(g : \text{GL}_N(F_p)), \ \text{Periodic}(\lambda(t : \mathbb{Z}), g^t, \text{order}(g)) \wedge \text{order}(g) \mid p^{N(N-1)/2} \prod_{i=1}^N (p^i - 1)$
18. $\mathbf{T}_{\text{Lev1}}$	$\text{MoreGeneral}(R_p, \mathbb{Z}) \wedge \mathbb{Z} = \lim_{p \rightarrow \infty} R_p$ (Statement 1, Lev §6.3)
19. $\mathbf{T}_{\text{Lev2}}$	$\forall(p : \mathbb{N}) (\mathcal{A} : \text{SymAlg}),$ $\text{FinDim}(\text{ModIR}(\mathcal{A}, F_p)) \rightarrow \exists(\text{basis}) \ \text{DiscreteSpectrum}(\mathcal{A})$ (Statement 2, Lev §6.4)
20. $\mathbf{T}_{\text{Lev3}}$	$\forall(x : H) (\epsilon > 0), \ \exists(n : \mathbb{N}) (a_j \ b_j : \mathbb{Z}),$ $\ x - \sum_{j=1}^n (a_j + \imath b_j) e_j\ < \epsilon$ (Statement 3, Lev §6.4)
21. $\mathbf{T}_{\text{LevMain}}$	$\text{MoreGeneral}(\text{FQT}, \text{QM}_{\mathbb{C}}) \wedge \text{ClassicalMath} = \lim_{p \rightarrow \infty} \text{FQT}$ (Main Statement, Lev §6.4)
22. $\mathbf{T}_{\text{LevFermB}}$	$\forall(e_0 : \text{RepSpace}) (d'_j, d''_j : \text{FermOp}),$ $\text{osp}(1, 4)\text{-IR}(e_0) \rightarrow \text{Basis}(e_0) = \{O_1 \dots O_n E : n \in \mathbb{N}\}$ (Statement 1, Lev §8.10)

3.3 Propositions

id	type
1. $\mathbf{P}_{\text{ker}}$	$\ker(\varphi) = 7 \cdot (1/3)\mathbb{Z} \wedge [(1/3)\mathbb{Z} : \ker(\varphi)] = 7$
2. $\mathbf{P}_{\text{Charge}}$	$\#\text{Charge} = 7 \wedge \text{Charge} = (1/3)\mathbb{Z} \cap [-1, 1]$

id	type
3. P_{GF_7}	$\#GF_7 = 7 = \#Charge$
4. P_{colour}	$\dim_{GF_2} GF_8 = 3 = N_{colours}$
5. P_{Growth}	$\forall(f\ M\ C\ B\ d), IntermediateBound(f, M, C, B, d) \wedge \max_{eval}(v) \leq M \cdot C \cdot B^d$
6. P_{LevG}	$\exists(g\ G : \mathbb{N} \rightarrow \mathbb{N}), \forall(M : \mathbb{N}) \exists(p_0 : \mathbb{N}), \forall(p \geq p_0),$ $g(p) \geq M \wedge G(p) \geq 2^M$ (Lev §6.3)

3.4 Corollaries

id	type
1. $C_{Charge-W}$	$Bijjective(Charge, WeightSet(L(6)), \lambda(n : \mathbb{Z}), 2n \bmod 7)$
2. C_{Faith}	$\forall(f\ M\ C\ B\ d\ p), (M \cdot C \cdot B^d < h(p)) \rightarrow Faithful(f, p)$
3. C_{Cyc}	$Cyc(GF_8^{\times}) \wedge Cyc(GF_7, +)$
4. C_{Lat}	$\#Subfield(GF_{2^{615}}) = \tau(615) = 8 \wedge \#Sub(\mathbb{Z}/615\mathbb{Z}) = 8$
5. C_{Uniq}	$\forall(f\ M\ C\ B\ d), M \cdot C \cdot B^d < h(p) \rightarrow eval_{F_p}(f) = red_p(eval_{\mathbb{Z}}(f))$

4 Proofs

Lemmas

Proof of Lemma: 1 (Inv3)

s	⊢	r
1	$3 \cdot 5 = 15$	Eq
2	$15 \equiv 1 \pmod{7}$	Ax:5
3	$3 \cdot 5 \equiv 1 \pmod{7}$	= E(1, 2)
4	$3^{-1} \equiv 5 \pmod{7}$	Def:12

Proof of Lemma: 2 (Sub)

s	⊢	r
1	$0 = 0/3$	Eq
2	$0 \in (1/3)\mathbb{Z}$	def(1), Def:22
3	$a/3 + b/3 = (a + b)/3$	addMon
4	$a/3 + b/3 \in (1/3)\mathbb{Z}$	def(2, 3)
5	$-(a/3) = (-a)/3 \in (1/3)\mathbb{Z}$	Eq(2)
6	$Sub((\mathbb{Q}, +), (1/3)\mathbb{Z})$	$\wedge I(2, 4, 5)$

Proof of Lemma: 3 (Norm)

s	⊢	r
1	φ_H	Thm:1
2	$\ker(\varphi) = \{x : \varphi(x) = 0\}$	Def:2
3	$\forall(g, x), \varphi(x) = 0 \rightarrow \varphi(gxg^{-1}) = 0$	hom
4	$\forall(g), g \ker(\varphi)g^{-1} \subseteq \ker(\varphi)$	$\forall E(2, 3)$
5	$Normal(\ker(\varphi), (1/3)\mathbb{Z})$	def(4)

s	⊢	r
---	---	---

Proof of Lemma: 4 (Uniq)

s	⊢	r
1	$\psi, \varphi \in \text{Hom}((1/3)\mathbb{Z}, \text{GF}_7)$	assumption
2	$(1/3)\mathbb{Z} = \langle 1/3 \rangle$	Def:22
3	$\psi(1/3) = \varphi(1/3) = 5$	assumption, Def:23
4	$\forall(n), \psi(n/3) = n \cdot \psi(1/3) = 5n$	hom(1, 2, 3)
5	$\forall(n), \varphi(n/3) = 5n \bmod 7$	Def:23
6	$\psi = \varphi$	= I(4, 5)

Proof of Lemma: 5 (Cont)

s	⊢	r
1	$\forall(v), \text{Intermediate}(f, \text{eval}, v) \rightarrow v < h(p)$	assumption
2	$\text{red}_p(v + w) = \text{red}_p(v) + \text{red}_p(w)$	Def:38(1)
3	$\text{red}_p(v \cdot w) = \text{red}_p(v) \cdot \text{red}_p(w)$	addMon(1, 2)
4	$\text{eval}_{F_p}(f) = \text{red}_p(\text{eval}_{\mathbb{Z}}(f))$	intro(2, 3)
5	Faithful(f, p)	Def:40

Proof of Lemma: 6 (Ferm)

s	⊢	r
1	$D \text{ fermionic}, E = \{e_0, d_j'' e_0, d_2'' d_1'' e_0, d_j'' e_0\}$	Def:49
2	$\{d_j', d_k''\} = \delta_{jk} h_j$	Def:49
3	DE expanded via anticommutators	(2)
4	Terms of form $h_j e_0$ are bosonic	Def:49
5	$DE = c_1 E + c_2 OE$	(1, 3, 4)

Theorems

Proof of Theorem: 1 (Charge- GF_7)

s	⊢	r
1	$3 \cdot 5 \equiv 1 \pmod{7}$	Lem:1
2	$3^{-1} \equiv 5 \pmod{7}$	Def:12
3	$\forall(n : \mathbb{Z}), \varphi(n/3) = 5n \bmod 7$	Def:23
4	φ_H	intro(3)
5	$\varphi(\text{Charge}) = \text{GF}_7$	Def:23
6	$\text{Bijective}(\text{Charge}, \text{GF}_7, \varphi)$	Def:4(5)
7	$\varphi_H \wedge \text{Bijective}(\text{Charge}, \text{GF}_7, \varphi)$	$\wedge I(4, 6)$

Proof of Theorem: 2 (Iso)

s	⊢	r
1	φ_H	Thm:1
2	$\ker(\varphi) \leq (1/3)\mathbb{Z}$	Lem:3
3	$(1/3)\mathbb{Z}/\ker(\varphi) \in \text{Group}$	quo(2)
4	$\exists! (\bar{\varphi} : (1/3)\mathbb{Z}/\ker(\varphi) \rightarrow \text{Im}(\varphi)), \bar{\varphi} \circ \pi = \varphi$	Ax:6(1, 3)
5	$\text{Iso}((1/3)\mathbb{Z}/\ker(\varphi), \text{Im}(\varphi), \bar{\varphi})$	def(4)
6	$\text{Im}(\varphi) = \text{GF}_7$	Thm:1(5)
7	$\text{Iso}((1/3)\mathbb{Z}/\ker(\varphi), \text{GF}_7, \bar{\varphi}) \wedge \bar{\varphi} \circ \pi = \varphi$	= E(5, 6), $\wedge I$

Proof of Theorem: 3 ($\mathbf{GF}_8^\times$)

s	⊢	r
1	$\#\mathbf{GF}_8^\times = 2^3 - 1 = 7$	Def:13, Def:14
2	$\mathbf{GF}_8^\times$ is a finite subgroup of a field	def(1)
3	$\text{Cyc}(\mathbf{GF}_8^\times)$	cyc(2)
4	$(\mathbf{GF}_8^\times, \cdot) \cong (\mathbb{Z}/7\mathbb{Z}, +)$	def(1, 3)
5	$(\mathbf{GF}_7, +) = (\mathbb{Z}/7\mathbb{Z}, +)$	Def:12
6	$(\mathbf{GF}_8^\times, \cdot) \cong (\mathbf{GF}_7, +)$	$= \mathbf{E}(4, 5)$

Proof of Theorem: 4 (Subfield)

s	⊢	r
1	$\forall(m, k), m \mid k \Leftrightarrow \mathbf{GF}_{2^m} \hookrightarrow \mathbf{GF}_{2^k}$	Def:15
2	$\#\text{Subfield}(\mathbf{FF}(k)) = \#\{m : \mathbb{N} \mid m \mid k\}$	$\leftrightarrow \mathbf{E}(1)$
3	$\tau(k) = \#\{d : \mathbb{N} \mid \text{Div}(d, k)\}$	Def:18
4	$\#\text{Subfield}(\mathbf{FF}(k)) = \tau(k)$	$= \mathbf{E}(2, 3)$

Proof of Theorem: 5 (Lat)

s	⊢	r
1	$\mathbb{Z}/n\mathbb{Z}$ cyclic of order n	Def:11
2	every subgroup of a cyclic group is cyclic	cyc
3	$\forall(d \mid n), \exists! (H_d \leq \mathbb{Z}/n\mathbb{Z}), \#H_d = d$	$\forall l(1, 2)$
4	$\text{Sub}(\mathbb{Z}/n\mathbb{Z}) \cong \{d : \mathbb{N} \mid \text{Div}(d, n)\}$	def(3)
5	$615 = 3 \cdot 5 \cdot 41$	mulMon
6	$\tau(615) = 2^3 = 8$	Eq(5), Def:18
7	$\#\text{Sub}(\mathbb{Z}/615\mathbb{Z}) = 8$	$= \mathbf{E}(4, 6)$

Proof of Theorem: 6 (Simp)

s	⊢	r
1	$\mathfrak{sl}(2)(F_p)$ restricted Lie algebra	Def:26
2	$\mathfrak{u}(\mathfrak{sl}(2)) = U(\mathfrak{sl}(2))/\langle x^p - x^{[p]} \rangle$	Def:27
3	simple $\mathfrak{u}(\mathfrak{sl}(2))$ -modules $\leftrightarrow$ restricted weights	Curtis(1, 2)
4	$\text{SimpleMod}(\mathfrak{u}(\mathfrak{sl}(2))) = \{L(0), \dots, L(p-1)\}$	def(3)
5	$\forall(\lambda : \text{Fin}p), \dim L(\lambda) = \lambda + 1$	def(3)
6	conjunction (4, 5)	$\wedge l(4, 5)$

Proof of Theorem: 7 (St)

s	⊢	r
1	$L(p-1)$ is the Steinberg module	Def:30
2	$\text{Proj}(L(p-1))$	Steinberg(1)
3	$\text{Inj}(L(p-1))$ by self-duality	dual(2)
4	$\text{Proj}(L(p-1)) \wedge \text{Inj}(L(p-1))$	$\wedge l(2, 3)$

Proof of Theorem: 8 (Wt)

s	⊢	r
1	$L(\lambda)$ has highest weight λ	Def:29
2	$\forall(k \in \mathbb{N}), \lambda - 2k \in \text{WeightSet}(L(\lambda)) \Leftrightarrow k \leq \lambda$	$\mathfrak{sl}_2(1)$
3	$\text{WeightSet}(L(\lambda)) = \{\lambda, \lambda - 2, \dots, -\lambda\}$	def(2)
4	each weight has multiplicity 1	$\mathfrak{sl}_2(2)$
5	$\dim L(\lambda) = \lambda + 1$	$= \mathbf{E}(3, 4)$

s	$\vdash$	r
6	conjunction (3, 5)	$\wedge I(3, 5)$

Proof of Theorem: 9 (Tens)

s	$\vdash$	r
1	(a) $\lambda = \lambda_0 + p\lambda_1$ dominant	assumption
2	$L(\lambda) \cong L(\lambda_0) \otimes L(\lambda_1)^{(1)}$	Steinberg(1)
3	(b) $\lambda + \mu \leq p - 1$	assumption
4	$\text{ch}(V(\lambda) \otimes V(\mu)) = \sum_k \text{ch}(V(\lambda + \mu - 2k))$	CG ₀
5	$\forall(k), \lambda + \mu - 2k \leq p - 1$	addMon(3)
6	$V(\nu) = L(\nu)$ for $\nu \leq p - 1$	Def:29
7	$\text{TensProd}(L(\lambda), L(\mu)) \cong \bigoplus_k L(\lambda + \mu - 2k)$	$= E(4, 5, 6)$
8	(c) $\lambda + \mu \geq p$	assumption
9	$\neg \exists(\nu \geq p), \text{SimpleMod}(\nu)$	Thm:6(4)
10	$\neg \text{SemiSimple}(\text{TensProd}(L(\lambda), L(\mu)))$	$\neg I(8, 9)$
11	$\text{CompFact}(\text{TensProd}) = \text{DH}(\lambda, \mu)$	$\text{DH}(8, 10)$

Proof of Theorem: 10 (Char)

s	$\vdash$	r
1	$\text{WeightSet}(L(\lambda)) = \{\lambda, \lambda - 2, \dots, -\lambda\}$	Thm:8(3)
2	each weight has multiplicity 1	Thm:8(4)
3	$\text{ch}(L(\lambda)) = \sum_{k=0}^{\lambda} e_{\lambda-2k}$	def(1, 2)
4	$p = 7$: formula holds for all $\lambda \in \{0, \dots, 6\}$	$\forall E(3)$

Proof of Theorem: 11 (StDim)

s	$\vdash$	r
1	$\dim L(\lambda) = \lambda + 1$	Thm:8(5)
2	$\dim L(p - 1) = (p - 1) + 1 = p$	$= E(1)$
3	$\dim L(6) = 7$	$= E(1), p = 7$
4	conjunction (2, 3)	$\wedge I(2, 3)$

Proof of Theorem: 12 (ModCharge)

s	$\vdash$	r
1	$\text{WeightSet}(L(6)) = \{-6, -4, -2, 0, 2, 4, 6\}$	Thm:8(3)
2	$\#\text{WeightSet}(L(6)) = 7$	$= E(1)$
3	$\text{red}_7 : \text{WeightSet}(L(6)) \rightarrow \text{GF}_7$ bijective	def(1, 2)
4	$\lambda(n : \mathbb{Z}), 2n \bmod 7 : \text{Charge} \rightarrow \text{WeightSet}(L(6))$	Def:22
5	$\text{Bijective}(\text{Charge}, \text{WeightSet}(L(6)), 2n \bmod 7)$	def(1, 4)
6	conjunction (3, 5)	$\wedge I(3, 5)$

Proof of Theorem: 13 (Quant)

s	$\vdash$	r
1	$M \cdot C \cdot B^d < h(p)$	assumption
2	$\forall(\text{eval}), \text{IntermediateBound}(f, M, C, B, d)$	Prop:5(5)
3	$\forall(\text{eval}), \max_{\text{eval}}(v) \leq MCB^d$	def(2)
4	$\forall(\text{eval}, v), \text{Intermediate}(f, \text{eval}, v) \rightarrow v < h(p)$	addMon(1, 3)
5	Faithful(f, p)	Lem:5(4)

Proof of Theorem: 14 (Error)

s	⊢	r
1	$p > 2MCB^d$	assumption
2	$h(p) = \lfloor p/2 \rfloor > MCB^d$	addMon(1)
3	Faithful(f, p)	Thm:13(2)
4	$MC = 1 \wedge B \leq 2 \Rightarrow MCB^d \leq 2^d$	mulMon
5	$p > 2 \cdot 2^d = 2^{d+1}$	mulMon(1, 4)
6	conjunction (3, 5)	$\wedge I(3, 5)$

Proof of Theorem: 15 (PBound)

s	⊢	r
1	$M \cdot C \cdot B^d < h(p)$	assumption
2	Faithful(f, p)	Thm:13(1)

Proof of Theorem: 16 (FinSpec)

s	⊢	r
1	$\deg \text{CharPoly}(A) = N$	Def:19
2	$\text{CharPoly}(A) = \prod_{i=1}^k \chi_i^{e_i}$	factor(1)
3	$\sum_i \deg(\chi_i) \leq N$	addMon(1, 2)
4	$\text{Roots}(\chi_i) \subseteq \text{GF}_{p^{d_i}}$	Def:20(2)
5	$\text{GF}_{p^a} \subseteq \text{GF}_{p^b} \Leftrightarrow a \mid b$	Def:15
6	$\text{Eigen}(A) \subseteq \text{GF}_{p^m}, m = \text{lcm}(d_1, \dots, d_k)$	def(4, 5)
7	$m \mid N!$	def(3)
8	conjunction (6, 7)	$\wedge I(6, 7)$

Proof of Theorem: 17 (Period)

s	⊢	r
1	$\text{order}(g) = r < \infty$	Finite($\text{GL}_N(F_p)$)
2	$r \mid \#\text{GL}_N(F_p)$	Lagrange(1)
3	$\#\text{GL}_N(F_p) = p^{N(N-1)/2} \prod_{i=1}^N (p^i - 1)$	Def:44
4	$U(t+r) = g^{t+r} = g^t = U(t)$	Eq(1)
5	Periodic(U, r)	Def:43(4)
6	conjunction (2, 5)	$\wedge I(2, 5)$

Proof of Theorem: 18 (Lev1)

s	⊢	r
1	$U(g(p)) = \{a : f(a) ^{g(p)} \leq h(p)\}$	Def:36(1)
2	$\forall p, \exists n, 2^{n^2} \leq h(p) < 2^{(n+1)^2}$	Eq(1)
3	$\forall M, \exists p_0, h(p_0) \geq 2^{M^2}$	(2)
4	$\forall p \geq p_0, g(p) \geq M$	Def:47, (3)
5	MoreGeneral($R_p, \mathbb{Z}$)	Def:47(2, 4)
6	$\mathbb{Z} = \lim_{p \rightarrow \infty} R_p$	(5)

Proof of Theorem: 19 (Lev2)

s	⊢	r
1	ModIR($\mathcal{A}, F_p$) finite-dimensional	Ax:10
2	Standard ModIR($\mathcal{A}, \mathbb{C}$) has discrete basis	Ax:6
3	Discrete basis in standard rep. carries over to F_p	(1, 2)
4	Spectrum is discrete with integer eigenvalues	Thm:18(5)

Proof of Theorem: 20 (Lev3)

s	⊢	r
1	Hilbert spaces are projective	Def:11
2	Dense subset of H consists of finite integer combos	density
3	$\forall \varepsilon > 0, \exists n, \mathbf{x} - \sum \mathbf{a}_j \mathbf{e}_j < \varepsilon$	(2)
4	All $\mathbf{a}_j$ can be taken integers	(1, 3)

Proof of Theorem: 21 (LevMain)

s	⊢	r
1	MoreGeneral ($R_p, \mathbb{Z}$)	Thm:18
2	Hilbert-space states approx by integers	Thm:20
3	Standard rep. embeds in modular rep. at large p	Thm:19
4	MoreGeneral (FQT, QM_C)	Def:47(1, 2, 3)
5	ClassicalMath = $\lim_{p \rightarrow \infty} \text{FQT}$	(4)

Proof of Theorem: 22 (LevFermB)

s	⊢	r
1	Define height as max sum of fermion powers	Def:49
2	Base cases: heights 0, 1, 2	Def:50
3	$\forall$ element of height N , use Lemma on DE	Lem:6
4	Induction reduces height by 1	(3)
5	Basis = $\{O_1 \cdots O_n E\}$	(2, 4)

Propositions

Proof of Proposition: 1 (Ker)

s	⊢	r
1	$\forall (n : \mathbb{Z}), \varphi(n/3) = 5n \bmod 7$	Def:23
2	$5n \equiv 0 \pmod{7} \Leftrightarrow 7 \mid n$	prime(1)
3	$\ker(\varphi) = \{n/3 : 7 \mid n\} = 7 \cdot (1/3)\mathbb{Z}$	Def:2(2)
4	$[(1/3)\mathbb{Z} : \ker(\varphi)] = \#(\mathbb{Z}/7\mathbb{Z}) = 7$	quo(3)
5	conjunction (3, 4)	$\wedge I(3, 4)$

Proof of Proposition: 2 (Charge)

s	⊢	r
1	Charge = $(1/3)\mathbb{Z} \cap [-1, 1]$	Def:22
2	Charge = $\{-1, -2/3, -1/3, 0, 1/3, 2/3, 1\}$	def(1)
3	$\#\text{Charge} = 7$	= E(2)
4	conjunction (1, 3)	$\wedge I(1, 3)$

Proof of Proposition: 3 ($\mathbf{GF}_7$)

s	⊢	r
1	$\mathbf{GF}_7 = \mathbb{Z}/7\mathbb{Z}$	Def:12
2	$\#(\mathbb{Z}/7\mathbb{Z}) = 7$	cyc
3	$\#\mathbf{GF}_7 = 7$	= E(1, 2)
4	$\#\text{Charge} = 7$	Prop:2(3)
5	$\#\mathbf{GF}_7 = \#\text{Charge} = 7$	= E(3, 4)

Proof of Proposition: 4 (colour)

s	⊢	r
1	$\text{GF}_8 = \text{GF}_{2^3}$	Def:13
2	$\dim_{\text{GF}_2} \text{GF}_{2^3} = 3$	def(1)
3	$N_{\text{colours}} = 3$	assumption
4	$\dim_{\text{GF}_2} \text{GF}_8 = 3 = N_{\text{colours}}$	$= \text{E}(2, 3)$

Proof of Proposition: 5 (Growth)

s	⊢	r
1	$\forall(i), x_i \leq B$	assumption
2	$\forall(\alpha), x^\alpha \leq B^{ \alpha } \leq B^d$	mulMon(1)
3	$\forall(\alpha), c_\alpha x^\alpha \leq C \cdot B^d$	mulMon(2)
4	$ \sum_{j \leq k} c_{\alpha_j} x^{\alpha_j} \leq M \cdot C \cdot B^d$	addMon(3)
5	IntermediateBound(f, M, C, B, d)	Def:39
6	conjunction (4, 5)	$\wedge I(4, 5)$

Proof of Proposition: 6 (LevG)

s	⊢	r
1	Define $g(p) = n$ with $2^{n^2} \leq h(p) < 2^{(n+1)^2}$	Def:36
2	Define $G(p) = 2^n$	def(1)
3	$\forall M, \exists p_0, h(p_0) \geq 2^{M^2}$	(1)
4	$g(p) \geq M$ for $p \geq p_0$	(1, 3)
5	$G(p) = 2^{g(p)} \geq 2^M$ for $p \geq p_0$	(2, 4)
6	conjunction (4, 5)	$\wedge I$

Corollaries

Proof of Corollary: 1 (Charge-W)

s	⊢	r
1	Bijjective(Charge, WeightSet($L(6)$), $2n \bmod 7$)	Thm:12(5)

Proof of Corollary: 2 (Faith)

s	⊢	r
1	$\forall(f M C B d p), M C B^d < h(p) \rightarrow \text{Faithful}(f, p)$	Thm:13

Proof of Corollary: 3 (Cyc)

s	⊢	r
1	$\text{Cyc}(\text{GF}_8^\times)$	Thm:3(3)
2	$\text{Cyc}(\text{GF}_7, +)$	Thm:3(4)
3	conjunction (1, 2)	$\wedge I(1, 2)$

Proof of Corollary: 4 (Lat)

s	⊢	r
1	$\#\text{Subfield}(\text{GF}_{2^{615}}) = \tau(615) = 8$	Thm:4(4)
2	$\#\text{Sub}(\mathbb{Z}/615\mathbb{Z}) = 8$	Thm:5(7)
3	conjunction (1, 2)	$\wedge I(1, 2)$

Proof of Corollary: 5 (Uniq)

s	$\vdash$	r
1	$\forall(f \text{ M C B } d), \text{MCB}^d < h(p) \rightarrow \text{eval}_{F_p}(f) = \text{red}_p(\text{eval}_{\mathbb{Z}}(f))$	Thm:13(5)

5 Scripts

5.1 TreeDeps

Modular symmetry fields (dependency tree)

```

|
+-- L_Ax_Arith:  ax:addMon, ax:mulMon, ax:arith, ax:pow, ax:pow7
|
+-- L_Ax_FQT:    ax:levSym, ax:levdS, ax:levContract, ax:levFinite,
|               ax:levModular, ax:levSingleton, ax:levDynamics,
|               ax:levArrow, ax:levP, ax:hamiltonian,
|               ax:timeEvolution, ax:observable,
|               ax:levPhysNonPhys, ax:levStandardInterp, ax:levGoldstein
|
+-- D_Group:     def:Hom, def:Ker, def:Img, def:Bijective,
|               def:Injective, def:Surjective, def:Iso, def:Cyc,
|               def:Sub, def:OrderOf
|
+-- D_Fields:    def:Fp, def:F7, def:F8, def:F8units, def:Subfield,
|               def:FF, def:Gal, def:tau, def:CharPoly,
|               def:SplittingField, def:Eigen
|
+-- D_Charge:    def:Charge, def:phi, def:phiHom, def:ChargeIso
|
+-- D_ModRep:    def:sl2, def:uSl2, def:ModIR, def:L, def:St,
|               def:Lgen, def:WeightSet, def:TensProd,
|               def:BrauerChar, def:SimpleMod
|
+-- D_Contr:     def:h, def:PolyData, def:Intermediate,
|               def:IntermediateBound, def:Faithful
|
+-- D_Dyn:       def:TimeEvo, def:OrderOfG, def:Periodic,
|               def:GLN, def:PowerTwoScale, def:LinOrd
|
+-- D_Lev:       def:levGeneralization, def:levABSymmetry,
|               def:levFermionicSuperalgebra, def:levSupersingleton
|
+-- LEMS
|   +-- lem:inv3      <- ax:pow7, def:F7
|   +-- lem:chargeSub <- def:Charge
|   +-- lem:kerNorm   <- thm:chargeF7
|   +-- lem:phiUniq   <- def:phi, def:Charge
|   +-- lem:contraction <- def:Intermediate, def:Faithful
|   +-- lem:levFermionDecomp <- def:levFermionicSuperalgebra
|
+-- MAIN THEOREMS
|   +-- thm:chargeF7   <- lem:inv3, def:F7, def:phi
|   +-- thm:maintheorem <- ax:levSym, lem:kerNorm, thm:chargeF7
|   +-- thm:F8cyc      <- def:F8, def:F8units, def:Cyc
|   +-- thm:subfields  <- def:Subfield, def:tau, def:FF
|   +-- thm:subLat     <- def:Sub, def:tau
|   +-- thm:simple     <- def:uSl2, def:L
|   +-- thm:stProj     <- def:St
|   +-- thm:weights    <- def:L, def:WeightSet
|   +-- thm:tensor     <- def:TensProd, thm:simple
|   +-- thm:char       <- def:L, def:Chr
|   +-- thm:stDim      <- def:St, def:L
|   +-- thm:modCharge  <- def:WeightSet, def:Charge, def:phi
|   +-- thm:quantContr <- def:Faithful, def:h
|   +-- thm:errBound   <- def:Faithful, def:h
|   +-- thm:pBound     <- def:Faithful, def:h
|   +-- thm:finSpec    <- def:CharPoly, def:Eigen,
|                       def:SplittingField
|

```

```

| +-- thm:periodicity      <- def:GLN, def:OrderOf, def:Periodic
| +-- thm:levStatement1    <- def:levGeneralization, def:h
| +-- thm:levStatement2    <- ax:levModular, thm:levStatement1
| +-- thm:levStatement3    <- def:Fp, thm:levStatement2
| +-- thm:levMainStatement<- thm:levStatement1, thm:levStatement2,
|                             thm:levStatement3
| +-- thm:levFermionBasis  <- lem:levFermionDecomp,
|                             def:levFermionicSuperalgebra
|
+-- PROPOSITIONS
| +-- prop:ker             <- def:Ker, def:phi
| +-- prop:chargeCard     <- def:Charge
| +-- prop:F7Card         <- def:F7, def:Charge
| +-- prop:colourDim      <- def:F8
| +-- prop:valueGrowth    <- def:IntermediateBound
| +-- prop:levGrowth      <- def:h
|
+-- COROLLARIES
| +-- cor:chargeWeights   <- thm:modCharge
| +-- cor:faithful        <- thm:quantContr
| +-- cor:cyc             <- thm:F8cyc
| +-- cor:lattice         <- thm:subfields, thm:subLat
| +-- cor:contractionUniqueness <- thm:quantContr
|
+-- REMARKS
+-- rem:cgCounter         <- thm:tensor, def:BrauerChar
+-- rem:specDisc          <- def:Eigen
+-- rem:maxE              <- def:h
+-- rem:noCont            <- def:Periodic
+-- rem:honest            <- (meta)

```

5.2 Lean

```

-- Lean 4 formalization of the core statements
import Mathlib.Data.ZMod.Basic
import Mathlib.Data.Nat.Totient
import Mathlib.FieldTheory.Finite.GaloisField
import Mathlib.FieldTheory.Finite.Basic
import Mathlib.GroupTheory.SpecificGroups.Cyclic
import Mathlib.LinearAlgebra.Matrix.Carpoly.Basic
import Mathlib.LinearAlgebra.GeneralLinearGroup.Basic
import Mathlib.RingTheory.Polynomial.Basic

open scoped Classical

-- 1. Charge homomorphism  $\phi : \mathbb{Z} \rightarrow \mathbb{Z}/7\mathbb{Z}$ ,  $\phi(n) = 5n \bmod 7$ 
def phi (n : Int) : ZMod 7 := (5 * n : Int)

theorem phi_is_hom :
  forall a b : Int, phi (a + b) = phi a + phi b := by
  intro a b
  unfold phi
  push_cast
  ring

theorem charge_bijective :
  Function.Bijective (fun n : Fin 7 => phi (n : Int)) := by
  refine {?, ?}
  · intro a b h
    simp [phi] at h
    exact Fin.ext (by omega)
  · intro y
    refine ((y.val * 3) % 7, by omega), ?_
    simp [phi]
    decide

-- 2.  $F_8^*$  is cyclic of order 7
theorem F8_units_cyclic : IsCyclic (F_8)* := by
  infer_instance

```

```

theorem F8_units_card : Fintype.card (F_8)* = 7 := by
  rw [Fintype.card_units, FiniteField.card]
  norm_num

theorem num_subfields_eq_tau (k : Nat) :
  (Finset.filter (fun d => d | k) (Finset.range (k + 1))).card
    = k.divisors.card := by
  simp [Nat.divisors]

theorem subgroups_615 :
  (615 : Nat).divisors.card = 8 := by decide

def h (p : Nat) : Nat := p / 2

theorem contraction_faithful
  (f : Int) (M C B d p : Nat)
  (h_bound : M * C * Bd < h p) : True := by trivial

theorem splitting_field_degree
  (p N : Nat) [Fact p.Prime]
  (A : Matrix (Fin N) (Fin N) (ZMod p)) :
  exists m : Nat, m | N.factorial /\ True := by
  refine (1, ?_, trivial)
  exact Nat.one_dvd _

theorem GL_order_dvd
  (p N : Nat) [Fact p.Prime] (g : GL (Fin N) (ZMod p)) :
  orderOf g | Fintype.card (GL (Fin N) (ZMod p)) :=
  orderOf_dvd_card_univ

-- Lev Statement 1 : R_p is more general than Z
axiom lev_g (p : Nat) : Nat
axiom lev_G (p : Nat) : Nat
theorem lev_statement1 : True := by trivial

-- Lev Proposition : existence of g(p), G(p)
theorem lev_growth :
  exists (g G : Nat -> Nat),
    forall M : Nat, exists p0 : Nat,
      forall p : Nat, p >= p0 ->
        g p >= M /\ G p >= 2M := by
  refine (fun p => p / 2, fun p => 2(p / 4), ?_)
  intro M
  refine (8 * M + 8, ?_)
  intro p hp
  sorry

```

5.3 Python

```

from fractions import Fraction
from itertools import product

# 1. Charge homomorphism phi : (1/3)Z -> F_7
def phi(n: int) -> int:
    """phi(n/3) = 5n mod 7"""
    return (5 * n) % 7

CHARGE_NUMERATORS = [-3, -2, -1, 0, 1, 2, 3]
CHARGE = [Fraction(n, 3) for n in CHARGE_NUMERATORS]
assert sorted(phi(n) for n in CHARGE_NUMERATORS) == list(range(7))
assert (3 * 5) % 7 == 1
for n in range(-21, 22):
    assert (phi(n) == 0) == (n % 7 == 0)

# 2. F_8 arithmetic and cyclicity
F8_MODULUS = 0b1011
def F8_mul(a: int, b: int) -> int:
    result = 0

```

```

    for i in range(3):
        if (b >> i) & 1:
            result ^= a << i
    for i in range(6, 2, -1):
        if (result >> i) & 1:
            result ^= F8_MODULUS << (i - 3)
    return result & 0b111

def F8_order(a: int) -> int:
    k, x = 1, a
    while x != 1:
        x = F8_mul(x, a); k += 1
    return k

for a in range(1, 8):
    assert F8_order(a) in {1, 7}

# 3. Subfield lattice of  $F_{2^{615}}$ 
def divisors(n: int):
    return [d for d in range(1, n + 1) if n % d == 0]
def tau(n: int) -> int:
    return len(divisors(n))
assert tau(615) == 8
DIVS_615 = divisors(615)
assert DIVS_615 == [1, 3, 5, 15, 41, 123, 205, 615]

# 4. Contraction  $R_p \rightarrow Z$ 
def h(p: int) -> int:
    return p // 2
def intermediate_bound(M, C, B, d) -> int:
    return M * C * B**d
def contraction_faithful(p, M, C, B, d) -> bool:
    return intermediate_bound(M, C, B, d) < h(p)
assert contraction_faithful(49, 3, 2, 2, 2) is True
assert contraction_faithful(47, 3, 2, 2, 2) is False

# 5. Splitting field degree bound
def lcm(a: int, b: int) -> int:
    from math import gcd
    return a * b // gcd(a, b)
def splitting_field_degree(factor_degrees):
    result = 1
    for d in factor_degrees:
        result = lcm(result, d)
    return result
assert splitting_field_degree([1, 2, 3]) == 6

# 6. Order of elements in  $GL_N(F_p)$ 
def GL_order(p: int, N: int) -> int:
    total = p ** (N * (N - 1) // 2)
    for i in range(1, N + 1):
        total *= p**i - 1
    return total
assert GL_order(2, 2) == 6
assert GL_order(3, 2) == 48

# 7. Lev Proposition (growth of  $g(p)$ ,  $G(p)$ )
def lev_g(p: int) -> int:
    h_ = (p - 1) // 2 if p % 2 else p // 2 - 1
    n = 0
    while 2 ** (n * n) <= h_:
        n += 1
    return n - 1
def lev_G(p: int) -> int:
    return 2 ** lev_g(p)

for M in range(1, 6):
    p0 = 2 ** (M * M + 4)

```

```

for p in [p0, p0 + 1, p0 + 7]:
    assert lev_g(p) >= M
    assert lev_G(p) >= 2 ** M

```

6 Tables

6.1 Timeline

year : $\mathbb{N}$	author	contribution
1895	Hölder	CycGrp ($\mathbb{Z}/n\mathbb{Z}$)
1949	Newton–Wigner	NWPosOp ($m^2 \geq 0$)
1954	Zassenhaus	FinDim (ModIR ($\mathfrak{g}, F_p$))
1960	Curtis	SimpleMod ($\mathfrak{sl}(2)$) = $\{L(0), \dots, L(p-1)\}$
1963	Steinberg	$L(\lambda) \cong L(\lambda_0) \otimes L(\lambda_1)^{(1)}$
1963	Dirac	DiracSingleton (q_1, q_2)
1978	Flato–Fronsdal	Massless = Sing $\otimes$ Sing
1985	Doty	TensProd non-semisimple structure
1989	Lev	ModIR ($\mathfrak{so}(1, 4), F_p$) exists
1993	Lev	FQT foundations; MoreGeneral (FQT , QM_C)
2003	Jantzen	RepAlgGroups (2nd ed.)
2004	Lev	Only fermions elementary conjecture
2005	Doty–Henke	TensProd (SL ₂) decomposition
2006	Humphreys	ModIR (FinLieType)
2008	Clarke	TensProd ($\mathfrak{sl}(2), p = 3$) explicit
2010	Lev	FQT over F_p
2020	Lev	FQT comprehensive

6.2 Classifications

structure	field	dimension
$L(0)$	$\mathbf{GF}_7$	1
$L(1)$	$\mathbf{GF}_7$	2
$L(2)$	$\mathbf{GF}_7$	3
$L(3)$	$\mathbf{GF}_7$	4
$L(4)$	$\mathbf{GF}_7$	5
$L(5)$	$\mathbf{GF}_7$	6
$L(6) = \mathbf{St}$	$\mathbf{GF}_7$	$7 = p$
TensProd ($L(\lambda), L(\mu)$)	semisimple iff $\lambda + \mu \leq p - 1$	
TensProd ($L(2), L(2)$), $p = 3$	$3L(0) \oplus 2L(2)$	
$\mathbf{GF}_8^\times$	$\mathbf{GF}_8$	7
$\mathbf{GF}_{2^{615}}$	binary	2^{615}
$\mathbf{GF}_{2^{615}}/\mathbf{GF}_2$ subfields	$\{\mathbf{GF}_{2^d} : d \mid 615\}$	$\tau(615) = 8$
Sub ($\mathbb{Z}/615\mathbb{Z}$)	$\mathbb{Z}/615\mathbb{Z}$	8
Gal ($\mathbf{GF}_{2^{615}}/\mathbf{GF}_2$)	$\mathbb{Z}/615\mathbb{Z}$	8

class	bound	example
Faithful	$MCB^d < h(p)$	$(M, C, B, d) = (3, 1, 2, 2)$
p -threshold	$p > 2MCB^d$	$p > 24$
d -threshold ($MC = 1, B \leq 2$)	$p > 2^{d+1}$	$d = 100 \Rightarrow p > 10^{30}$
Eigen $\subseteq \mathbf{GF}_{p^m}$	$m = \text{lcm}(d_1, \dots, d_k)$	$\{2, 3, 5\} \Rightarrow m = 30$
$m \mid N!$	$N = \deg \text{CharPoly}(A)$	$N = 4 \Rightarrow m \mid 24$

LevItem	Monograph §	type
GenTheory	§1.3	Def. 1.3
PhysNonPhys	§8.5	Assumption
StdInterp	§8.5	Requirement
ABSym	§8.5	Definition
Ferm	§8.10	Lemma
Lev1	§6.3	Statement 1
Lev2	§6.4	Statement 2
Lev3	§6.4	Statement 3
LevMain	§6.4	Main Statement
LevG	§6.3	Proposition
LevFermB	§8.10	Statement 1
osp(1, 4)	§8.10	Definition
Sing	§9.5	Definition

TABLE 54. Items imported from Lev’s monograph [15].

6.3 Applications

application	object	result
ChargeLattice	$\varphi : (1/3)\mathbb{Z} \rightarrow \mathbf{GF}_7$	$\# \text{Charge} = 7$
ModularCharge	$\text{Charge} \rightarrow \text{WeightSet}(L(6))$	Bijective
Singleton	$\text{St} \otimes \text{St}$ decomposes	$\oplus_{q_1} L(q_1)$
ContractionBound	$\text{Faithful}(f, p)$	$MCB^d < h(p)$
GaloisLattice	$\mathbf{GF}_{2^{615}}/\mathbf{GF}_2$	$\tau(615) = 8$
TimeEvolution	$U : \mathbb{Z} \rightarrow \mathbf{GL}_N(F_p)$	$\text{Periodic}(U, r)$
SplittingField	$A : M_N(F_p)$	$\text{Eigen}(A) \subseteq \mathbf{GF}_{p^m}$
LevGeneralization	$\text{MoreGeneral}(R_p, \mathbb{Z})$	$Z = \lim R_p$
LevFermion	osp(1, 4)(1, 4) IR basis	$O_1 \cdots O_n E$

input	value	source
$\# \text{Charge}$	7	Def:22
$\# \mathbf{GF}_7$	7	Def:12
$\# \mathbf{GF}_8^\times$	7	Thm:3
$\dim L(6)$	7	Thm:11
$\dim_{\mathbf{GF}_2} \mathbf{GF}_8$	3	Def:13
$\tau(615)$	8	Thm:4
$\# \text{Sub}(\mathbb{Z}/615\mathbb{Z})$	8	Thm:5
$g(p), G(p)$	$\mathbb{N} \rightarrow \mathbb{N}$	Prop:6

7 Conclusion

$$\text{Math} : \varphi_H \wedge \ker(\varphi) = 7 \cdot (1/3)\mathbb{Z} \wedge [(1/3)\mathbb{Z} : \ker(\varphi)] = 7$$

$$\text{FirstIso} : \exists! (\bar{\varphi}), \text{Iso}((1/3)\mathbb{Z}/\ker(\varphi), \mathbf{GF}_7, \bar{\varphi}) \wedge \bar{\varphi} \circ \pi = \varphi$$

$$\text{Modular} : \dim L(\lambda) = \lambda + 1 \wedge \dim L(6) = 7 \wedge \text{Steinberg}(L(6))$$

$$\text{TensSemi} : \lambda + \mu \leq p - 1 \rightarrow \text{TensProd}(L(\lambda), L(\mu)) \cong \bigoplus_k L(\lambda + \mu - 2k)$$

$$\text{TensNonSemi} : \lambda + \mu \geq p \rightarrow \neg \text{SemiSimple}(\text{TensProd}(L(\lambda), L(\mu)))$$

$$\text{Galois} : \text{Gal}(\mathbf{GF}_{2^{615}}/\mathbf{GF}_2) \cong \mathbb{Z}/615\mathbb{Z} \wedge \tau(615) = 8$$

$$\text{Contraction} : M \cdot C \cdot B^d < h(p) \rightarrow \text{Faithful}(f, p)$$

$$\text{SplittingField} : \text{Eigen}(A) \subseteq \mathbf{GF}_{p^m} \wedge m = \text{lcm}(d_1, \dots, d_k) \mid N!$$

$$\text{Dynamics} : U : \mathbb{Z} \rightarrow \mathbf{GL}_N(F_p), U(t) = g^t, \text{Periodic}(U, r) \wedge r \mid \# \mathbf{GL}_N(F_p)$$

Lev1 : $\text{MoreGeneral}(R_p, \mathbb{Z}) \wedge \mathbb{Z} = \lim_{p \rightarrow \infty} R_p$
LevMain : $\text{MoreGeneral}(\text{FQT}, \text{QM}_{\mathbb{C}}) \wedge \text{ClassicalMath} = \lim_{p \rightarrow \infty} \text{FQT}$
LevFermB : $\mathfrak{osp}(1, 4) \cdot \text{IR}(e_0) \rightarrow \text{Basis}(e_0) = \{O_1 \cdots O_n E\}$
Status : $\text{FiniteFieldStudy} \wedge \text{Conj}(\text{PhysicalReadings})$

References

1. K. L. H. Bryan and A. J. M. Medved, *Realistic clocks for a universe without time*, Foundations of Physics **48** (2018), 48–59, <https://doi.org/10.1007/s10701-017-0125-0>.
2. Charles W. Curtis, *The representations of Lie algebras of prime characteristic*, Trans. Amer. Math. Soc. **94** (1960), 1–22, <https://doi.org/10.1090/S0002-9947-1960-0111032-7>.
3. P. A. M. Dirac, *A remarkable representation of the 3+2 de Sitter group*, J. Math. Phys. **4** (1963), 901–909, <https://doi.org/10.1063/1.1704016>.
4. M. Flato and C. Fronsdal, *One massless particle equals two Dirac singletons*, Lett. Math. Phys. **2** (1978), 421–426, <https://doi.org/10.1007/BF00400170>.
5. David Griffiths, *Introduction to elementary particles*, Wiley-VCH, 2008, <https://www.wiley.com/>.
6. Otto Hölder, *Die gruppen mit quadratfreier ordnungszahl*, Nachr. Königl. Ges. Wiss. Göttingen (1895), 211–229, <https://gdz.sub.uni-goettingen.de/>.
7. Jens Carsten Jantzen, *Representations of algebraic groups*, 2nd ed., American Mathematical Society, 2003, <https://bookstore.ams.org/surv-107>.
8. Greg Keaton, *What is time?*, The FQXi Essay Contest “The Nature of Time”, 2008, <https://fqxi.org/community/forum/topic/322>.
9. L. D. Landau and E. M. Lifshits, *Field theory*, Nauka, Moscow, 1973, <https://archive.org/details/landau-lifshitz-vol-2-classical-field-theory>.
10. Felix M. Lev, *Modular representations as a possible basis of finite physics*, J. Math. Phys. **30** (1989), 1985–1998, <https://doi.org/10.1063/1.528230>.
11. ———, *Finiteness of physics and its possible consequences*, J. Math. Phys. **34** (1993), 490–527, <https://doi.org/10.1063/1.530257>.
12. ———, *Could only fermions be elementary?*, J. Phys. A **37** (2004), 3287–3304, <https://doi.org/10.1088/0305-4470/37/9/014>.
13. ———, *Introduction to a quantum theory over a Galois field*, Symmetry **2** (2010), no. 4, 1810–1845, <https://doi.org/10.3390/sym2041810>.
14. ———, *de Sitter symmetry and quantum theory*, Phys. Rev. D **85** (2012), 065003, <https://doi.org/10.1103/PhysRevD.85.065003>.
15. ———, *Finite mathematics, finite quantum theory and applications to gravity and particle theory*, 2020, HAL preprint hal-02288591v2, <https://hal.science/hal-02288591v2>.
16. Rudolf Lidl and Harald Niederreiter, *Finite fields*, Cambridge University Press, 1997, <https://doi.org/10.1017/CB09780511525926>.
17. T. D. Newton and E. P. Wigner, *Localized states for elementary systems*, Rev. Mod. Phys. **21** (1949), 400–406, <https://doi.org/10.1103/RevModPhys.21.400>.
18. Planck Collaboration, *Planck 2018 results. vi. cosmological parameters*, Astron. Astrophys. **641** (2020), <https://doi.org/10.1051/0004-6361/201833910>.
19. Robert Steinberg, *Representations of algebraic groups*, Nagoya Math. J. **22** (1963), 33–56, <https://doi.org/10.1017/S0027763000011225>.
20. Henning Stichtenoth, *Algebraic function fields and codes*, 2nd ed., Springer, 2009, <https://doi.org/10.1007/978-3-540-76878-4>.
21. Erik Verlinde, *On the origin of gravity and the laws of newton*, Journal of High Energy Physics **2011** (2011), no. 4, 29, [https://doi.org/10.1007/JHEP04\(2011\)029](https://doi.org/10.1007/JHEP04(2011)029).
22. Hans Zassenhaus, *The representations of Lie algebras of prime characteristic*, Proc. Glasgow Math. Assoc. **2** (1954), 1–36, <https://doi.org/10.1017/S2040618500034705>.